PM10 Air Quality Forecasting

Time Series Analysis: Los Angeles, 2000–2022

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Data Source: EPA AirData Download Daily Data · Station: Los Angeles, CA · Monthly PM10 ($\mu g/m^3$)

Dataset & Preprocessing

Source: EPA AirData Download Daily Data

- Station: **60370002,Los Angeles**
- Pollutant: **PM10** (μg/m^3)
- Raw cadence: every **6 days** (not daily)
- Raw range: **January 2000 – September 2022**

Why compress to monthly?

EPA's PM10 readings at this station arrive on a **6-day cycle** (every 6th day is sampled), not every day. This creates **irregular gaps**. Most time series models require a **uniform, evenly-spaced** grid. Averaging all readings within each calendar month gives exactly **12 values per year**, removes the irregularity, reduces noise, and makes the seasonal structure cleaner.

Final Datasets

Split	**Period**	**Points**
Train	Jan 2000 – Dec 2015	**192 months**
Test	Jan 2016 – Sep 2022	**81 months**

Descriptive Statistics

	Train	**Test**
Mean (μg/m^3)	37.2	33.2
Std (μg/m^3)	13.0	13.2

The test mean is slightly lower (33.2 vs 37.2),

Seasonal Decomposition

What is decomposition?

Any observed monthly time series can be split into three additive layers:

$$y_t = \underbrace{T_t}_{\text{Trend}} + \underbrace{S_t}_{\text{Seasonal}} + \underbrace{R_t}_{\text{Residual}}$$

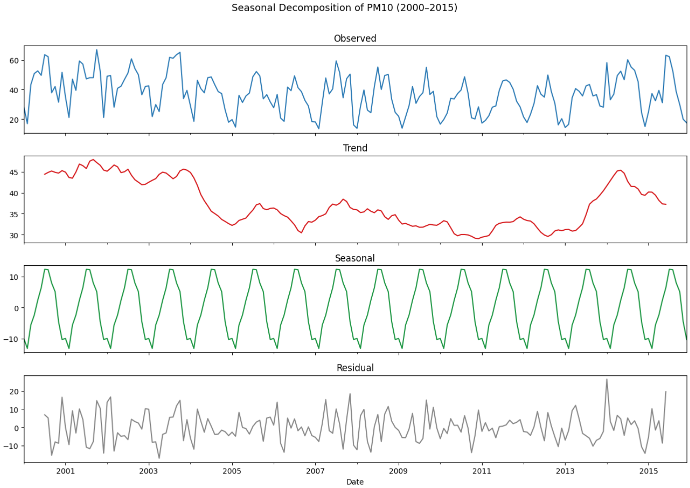
- **Trend** T_t: slow multi-year drift estimated via a centred 12-month moving average
- **Seasonal** S_t: repeating annual pattern — average of all same-month values after trend removal; tiles every 12 months
- **Residual** $R_t = y_t - T_t - S_t$: everything left; should look like random noise if the other two are well captured

What each component tells

Panel	What it tells you
Observed	Raw monthly PM10; overall shape
Trend	Mild **downward drift** 2000–2010 (emission regs), then flat
Seasonal	Clear **summer peak** (heat, stagnant air, wildfires) and **winter trough** (rain)
Residual	Random scatter — confirms decomposition captured structure

If the residual still showed a wave pattern, it would mean the seasonal estimate was wrong. Flat scatter = good decomposition.

Seasonal Decomposition: Plot



Key takeaways: (1) Trend shows LA's long-term improvement in air quality. (2) Seasonal panel confirms a consistent annual cycle with summer highs. (3) Residual is unstructured noise — decomposition is successful. This motivates using models with a **seasonal component** (SARIMA, Prophet).

Step 2 — Stationarity & the ADF Test

Augmented Dickey-Fuller (ADF) test:

- H_0: unit root exists ,which means series is non stationary
- H_1: no unit root which ,means series is stationary
- Reject H_0 if $p < 0.05$,which means that if the p value of the hypothesis test is less than 0.05 we reject H_0 and declare the series stationary

ADF Results

Series	***p*-value**	**Decision**
Original	0.2952	× Non-stationary
Seasonal diff	**0.0001**	✓ Stationary

Critical insight

The original series is non-stationary due to its **seasonal cycle**, not a linear trend. Regular differencing $(y_t - y_{t-1})$ removes trend but **leaves the seasonal cycle intact**. Seasonal differencing $(y_t - y_{t-12})$ is what makes it stationary.

ACF & PACF: Model Identification

Reading the plots

Pattern	Model
PACF cuts at p, ACF decays	AR(p)
ACF cuts at q, PACF decays	MA(q)
Both decay gradually	ARMA(p, q)
Spikes at lags 12, 24	Seasonal terms needed

Applied to seasonally-differenced PM10:

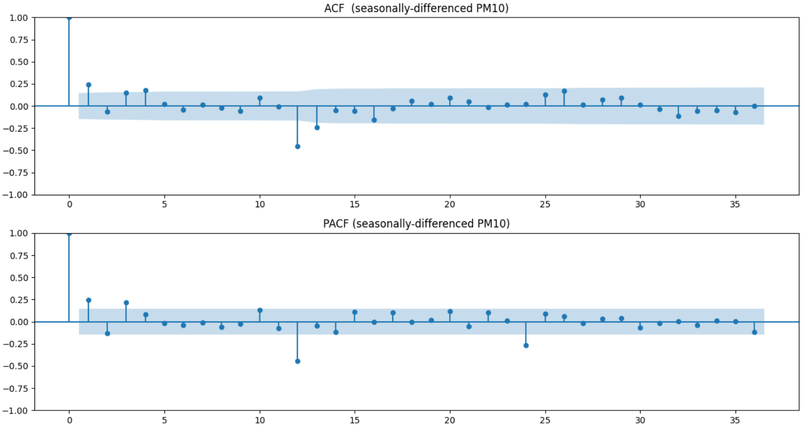
The blue shaded band is $\pm 1.96/\sqrt{n}$ — bars outside it are statistically significant autocorrelations.

What we see in our plots

- **Significant spikes at lags 12 and 24** in both ACF and PACF
  $\Rightarrow$ seasonal AR and MA terms needed: $P \geq 1$, $Q \geq 1$
- **ACF decays slowly** in first few lags
  $\Rightarrow$ MA terms in non-seasonal part ($q \geq 1$)
- **PACF spike at lag 1** then drops
  $\Rightarrow$ AR term in non-seasonal part ($p \geq 1$)

Conclusion: The data points towards SARIMA models with $(p, 0, q)(P, 1, Q)_{12}$ structure.

ACF & PACF: Plot (Seasonally-Differenced Series)



Key observations: Strong significant bars at lag 12 (and 24) in both ACF and PACF confirm a seasonal $s = 12$ structure. First few lags in PACF suggest $p = 1$ or $p = 2$. Slow ACF decay in early lags suggests $q = 1$ or $q = 2$. This directly guided the candidate model list in the SARIMA grid search.

SARIMA: Structure & Notation

SARIMA$(p, d, q)(P, D, Q)_{12}$

Six parameters controlling two layers of the model:

Symbol	Meaning	Best model
p	Non-seasonal AR order	1
d	Regular differencing	0
q	Non-seasonal MA order	2
P	Seasonal AR order	1
D	Seasonal differencing	1
Q	Seasonal MA order	1
s	Period	12

Full equation (using backshift operator B, $By_t = y_{t-1}$):

$$\underbrace{(1-\phi_1 B)(1-\Phi_1 B^{12})}\,\underbrace{(1-B^{12})}\,y_t$$

Why $d = 0$?

Seasonal differencing alone ($D = 1$) was sufficient to achieve stationarity (ADF $p = 0.0000$). Adding regular differencing would over-difference

SARIMA Grid Search: All Results

Model	AIC	RMSE	R^2
SARIMA(1,0,2)(1,1,1,12)	**1192.6**	**9.83**	**0.440**
SARIMA(2,0,2)(1,1,1,12)	1194.5	9.84	0.439
SARIMA(1,0,1)(1,1,1,12)	1214.6	9.94	0.427
SARIMA(1,0,1)(0,1,1,12)	1212.6	9.94	0.427
SARIMA(2,0,0)(0,1,1,12)	1228.4	10.59	0.350
SARIMA(2,0,1)(1,1,1,12)	1218.9	10.76	0.328
SARIMA(0,0,1)(0,1,1,12)	1215.7	11.07	0.289
SARIMA(0,0,0)(1,1,1,12)	1259.6	12.93	0.030
SARIMA(1,0,0)(1,1,0,12)	1255.2	15.30	−0.358
SARIMA(1,0,1)(1,1,0,12)	1254.4	15.37	−0.371

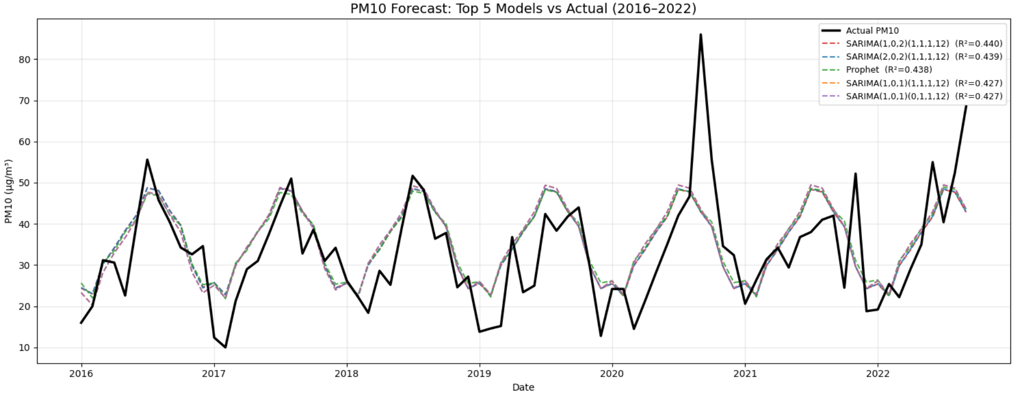
How the grid search works

1. ACF/PACF suggest plausible (p, q, P, Q) ranges
2. Fit each candidate SARIMA on **training data only**
3. Record AIC (penalised training fit)
4. Forecast **test set** (81 months)
5. Rank by **test RMSE** — the true measure

Why some models fail badly

Models with $Q = 0$ (no seasonal MA term) — e.g., $(1, 1, 0)_{12}$ — gave $R^2 < 0$, meaning they're *worse than predicting the mean*. ACF clearly showed a seasonal MA structure was needed; ignoring it is fatal.

Forecast Comparison: All Models vs. Actual

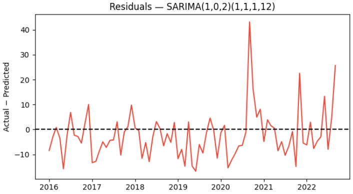


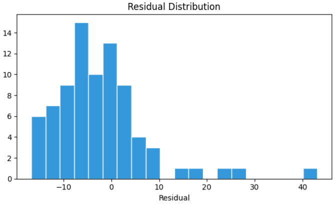
Key observations: SARIMA(1,0,2)(1,1,1,12) and Prophet track the seasonal oscillation correctly. XGBoost and LSTM show larger deviations, especially after 2019. All models under-predict the anomalous 2020 spike (California wildfire season) — this is expected: no model trained on 2000–2015 can anticipate an unprecedented event.

Full Model Comparison Table

Model	MAE	RMSE	MAPE (%)	R^2
SARIMA(1,0,2)(1,1,1,12)	**7.26**	**9.83**	**27.06**	**0.440**
SARIMA(2,0,2)(1,1,1,12)	7.28	9.84	27.15	0.439
Prophet	7.27	9.85	27.46	0.438
SARIMA(1,0,1)(1,1,1,12)	7.41	9.94	27.49	0.427
SARIMA(2,0,0)(0,1,1,12)	8.12	10.59	30.23	0.350
SARIMA(0,0,1)(0,1,1,12)	8.62	11.07	32.13	0.289
LSTM	8.24	11.37	29.43	0.251
XGBoost	8.78	11.88	33.08	0.181

Residual Diagnostics





What the plots tell us

The time plot shows residuals oscillating around zero with no significant wave pattern and the histogram is somewhat normal and there is left shift beacuse the model is overpredicting here as the test mean was lower than the train mean— the seasonal component was almost captured. Larger residuals around 2020 are expected: the unprecedented California wildfire season produced PM10 spikes that no model trained on 2000–2015 could anticipate. This is **irreducible error**, not a model flaw.

What are residuals?

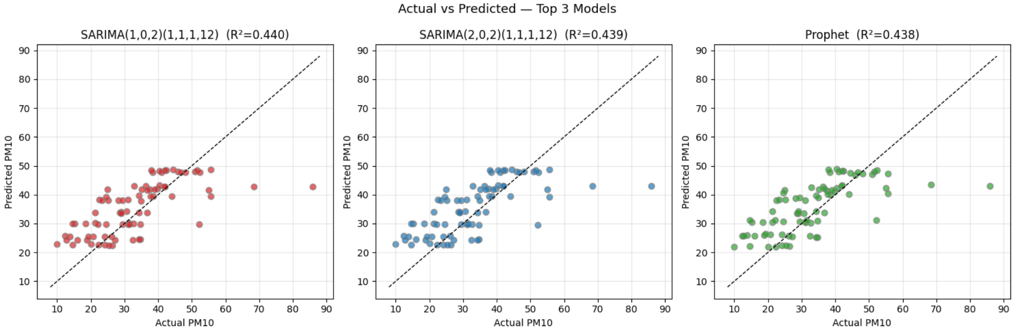
$$e_t = y_t - \hat{y}_t$$

The difference between what actually happene and what the model predicted.

What we want to see:

- Scattered randomly around zero — no leftover pattern
- Roughly constant spread over time (no growing errors)
- residual histogram following a bell curve centered at zero

Actual vs. Predicted (Top 3 Models)



How to read: Points on the dashed diagonal = perfect prediction. Vertical spread = error magnitude. **SARIMA** and **Prophet** show the tightest cloud around the diagonal — consistent with their $R^2 \approx 0.44$. The slight compression of predictions toward the mean (cloud slope $<45^\circ$) is normal in multi-step forecasting; models hedge under uncertainty. **LSTM** shows wider scatter but no systematic direction bias.

2-Year Test

Experiment: Re-run the entire pipeline but test on only **24 months** (2016–2017) instead of 81 months (2016–2022).

Previously the series converged to its seasonal pattern and also anomaly of an year(2020). R^2 should rise here substantially because the model hasn't yet converged to the mean seasonal pattern and still has real conditioning power to predict recent data.

2-Year Test Results (top models):

Model	RMSE	R^2
SARIMA(1,0,1)(1,1,1,12)	**7.11**	**0.591**
Prophet	7.15	0.586
SARIMA(1,0,2)(1,1,1,12)	7.33	0.565
XGBoost	7.63	0.529

Head-to-Head: 2-Year vs. 6.75-Year Horizon

Metric	81 months	24 months
Best R^2	0.440	**0.591**
Best RMSE	9.83	**7.11**
Best MAE	7.26	**5.78**

Conclusion: The models are working correctly. The low R^2 at 81 months is a *convergence effect*, not a model failure.

Conclusion

Summary of the pipeline:

1. **Data**: EPA AirData, LA station, 2000–2022. Irregular 6-day readings compressed to 12 uniform monthly values per year for model compatibility.
2. **Decomposition**: confirmed strong seasonal structure ($F = 0.55$), mild downward trend, and clean residuals.
3. **Stationarity**: ADF showed seasonal differencing ($D = 1$) sufficient; regular differencing not needed.
4. **ACF/PACF**: identified seasonal AR and MA terms at lag 12, guiding SARIMA candidate list.
5. **Grid search**: 10 SARIMA variants; best = $\mathbf{(1,0,2)(1,1,1)}_{12}$ at 81 months.
6. **Horizon effect**: R^2 rises from 0.44 → 0.59 at 24-month horizon — models are correct, long-horizon convergence is the cause.

Results Summary

Metric	81-mo	24-mo
Best model	SARIMA family	
RMSE (μg/m^3)	9.83	**7.11**
MAE (μg/m^3)	7.26	**5.78**
R^2	0.440	**0.591**

Potential improvements

- Add weather regressors (SARIMAX) $\Rightarrow$ expected $R^2 \approx 0.65$–0.75
- Hybrid: SARIMA residuals + XGBoost $\Rightarrow$ decomposition ensemble
- Neural-prophet or temporal fusion transformers with longer daily resolution